Nuclei Segmentation in Whole Slide Images Using U-Net Architecture

Abstract—**Cell nucleus segmentation is a fundamental task in digital pathology, enabling downstream applications such as tumor grading, morphological feature extraction, and biomarker quantification. Whole Slide Images (WSIs) present unique challenges for automated segmentation due to their extremely large size, staining variability, and the complex morphology of nuclei. In this project, we implemented and trained a U-Net model for cell nucleus segmentation. While the original plan involved manual segmentation of raw WSIs, the time and storage constraints associated with tiling and annotation made this approach infeasible. Instead, publicly available pre-segmented datasets were employed for training. We explored alternative approaches such as Meta's Segment Anything Model (SAM), but domain mismatches led to incorrect ground truth masks. Our U-Net model achieved a validation Dice coefficient of 0.91, demonstrating strong segmentation performance on the benchmark dataset. This paper details the methodology, including data preprocessing, model architecture, training strategies, and evaluation metrics. We also discuss the problems encountered during development, limitations of the current approach, and future improvements such as WSI tiling pipelines, advanced architectures, and post-processing methods for instance-level segmentation.**

Index Terms—Nuclei segmentation, whole slide images, digital pathology, U-Net, deep learning.

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I. Introduction

Histopathology is a critical tool in clinical diagnostics and cancer research, where microscopic examination of tissue slides provides insight into disease progression and treatment response. A key step in computational pathology is nuclei segmentation, the process of identifying and delineating individual cell nuclei within histological images. Accurate segmentation enables downstream tasks such as quantification of nuclear density, morphological analysis, and spatial modeling of tissue microenvironments.

Whole Slide Images (WSIs), which are digitized scans of entire pathology slides, are extremely high resolution, often reaching gigapixel scales. Analyzing WSIs introduces computational challenges due to their size, as well as biological challenges such as variation in nuclear morphology, irregular boundaries, and overlapping nuclei.

Deep learning has emerged as a powerful tool for biomedical image analysis. Among various architectures, U-Net has become the standard for segmentation in medical imaging due to its encoder–decoder design with skip connections, which enables precise localization while capturing contextual information.

In this project, we implement a U-Net model for nuclei segmentation. We initially planned to generate training data from raw WSIs through manual annotation, but the impracticality of this approach led us to adopt publicly available pre-segmented datasets instead. Despite this limitation, our experiments yielded competitive results, and the project highlights important lessons in dataset preparation, model development, and practical constraints in WSI analysis.

The model was trained on the 2018 Data Science Bowl dataset [1], which provides annotated images and masks of cell nuclei. Online tutorials [2] and videos [4] were also valuable in learning the U-Net architecture and implementing the model.

II. Related Work

Traditional methods for nuclei segmentation relied on handcrafted features, thresholding, and morphological operations. These methods, while simple, struggle with staining variability and overlapping nuclei.

With the rise of deep learning, convolutional neural networks (CNNs) revolutionized biomedical segmentation. The U-Net architecture, introduced by Ronneberger et al., has become particularly influential due to its ability to combine semantic information from deep layers with fine localization from shallow layers. Variants such as U-Net++, Attention U-Net, and residual U-Net further refine performance.

More recent approaches incorporate instance segmentation, where each nucleus is treated as a separate object. Models such as Mask R-CNN and HoVer-Net excel in this domain, separating touching nuclei using boundary predictions or object proposals.

Foundation models like Segment Anything (SAM) offer general-purpose segmentation capabilities. However, due to domain gaps, SAM often fails to produce accurate nuclei boundaries in histopathology images without specialized fine-tuning.

This project focuses on implementing a reproducible U-Net baseline, serving as a foundation for future work with more advanced models.

III. Methodology

This section describes in detail the methodological choices made during the development of the cell nucleus segmentation model. Each part of the pipeline is explained, with justifications for coding decisions, their advantages, and potential alternatives or improvements.

A. Data Ingestion and Preprocessing

The raw dataset was provided as compressed archives, which were extracted into the Colab environment under /content/working/. This setup allowed easy access to the dataset's folder structure, containing images and their corresponding mask directories. Handling the archive in this way simplified reproducibility across different machines, as Google Colab resets its filesystem between sessions.

1) The load_sample Function:: A central function in the pipeline was load_sample(image_id, target_size=(256,256)). It reads the RGB image using OpenCV and resizes it to the target size. It also iterates through all mask files belonging to that image (each mask corresponds to one nucleus). Lastly, it resizes each mask and merges them using np.maximum, resulting in a single binary mask with pixel values of 0,1.

The dataset provides instance masks (one per nucleus). For a baseline segmentation task, these were merged into a single binary semantic mask (nucleus vs. background). This reduced the complexity of the problem from instance segmentation to semantic segmentation.

The advantages are that it simplifies the modeling task, making U-Net directly applicable. Has faster training, since the network only predicts two classes and is memory-efficient, as only a single binary mask per image needs to be stored.

However, this merging approach loses information about individual nuclei. When two nuclei touch or overlap, they become indistinguishable in the binary mask. This explains why the trained model occasionally merged neighboring nuclei. Future work should incorporate instance-level labels or boundary detection to resolve this issue.

2) The data_loader Function: Another key function, data_loader(data_dir), collected all image IDs in the dataset directory, applied load_sample to each, and stored the resulting arrays as float32.

For datasets of moderate size (e.g., 10k images), eager materialization into NumPy arrays is feasible. This makes it easy to interface with TensorFlow datasets and avoids repeated disk reads during training.

As for possible improvements, normalization of inputs (X/255.0) would stabilize training by scaling intensities to [0,1]. Large-scale WSI datasets would require lazy loading or TFRecords for efficiency.

3) Data Splitting: The dataset was split into training and validation sets using an 80/20 ratio with train_test_split. This ensures that the model generalizes to unseen data and supports early stopping based on validation loss.

4) TensorFlow Data Pipeline: The training and validation datasets were then converted into TensorFlow tf.data.Dataset objects. This allowed for efficient batching, shuffling, and prefetching, ensuring smooth data flow during model training.

To prepare the data for training, the following steps were applied:

batch(16): groups data into batches of 16, balancing gradient stability with GPU memory usage.

shuffle: randomizes the order of training data to prevent the model from memorizing sequence patterns.

prefetch(AUTOTUNE): overlaps data preprocessing and model training to improve throughput.

The advantage is that TensorFlow's tf.data API provides pipeline parallelism, enabling efficient GPU utilization.

Future extensions could include integrating map() transformations for on-the-fly data augmentation (flips, rotations, brightness shifts, elastic deformations). This would improve robustness to staining and shape variations common in histopathology.

B. Model Architecture (U-Net)

The model used for this project was based on the U-Net architecture, which is particularly well suited for biomedical image segmentation. U-Net is composed of two main paths: an encoder that compresses the input image into a latent representation, and a decoder that reconstructs a dense segmentation map at the original resolution. The encoder extracts semantic features by progressively reducing the spatial resolution of the image, while the decoder leverages these representations to produce accurate localization of nuclei. A defining characteristic of U-Net is the presence of skip connections, which directly transfer feature maps from the encoder to the corresponding decoder layers, thereby preserving fine structural details.

1) Encoder: The encoder in this implementation was constructed using a series of convolutional blocks. Each block consisted of two successive convolutional layers with 3×3 kernels, followed by ReLU activation. Employing two convolutional layers per block increases the expressive capacity of the network while ensuring that local features are effectively captured. After each block, max pooling was used to reduce the spatial resolution by a factor of two, allowing the network to gain progressively larger receptive fields and capture contextual information that spans beyond individual nuclei.

2) Bottleneck: At the center of the network lies the bottleneck, which represents the most compressed form of the input image. This stage used 1024 filters, encoding high-level abstractions that differentiate nuclei from the surrounding tissue background. Although spatial information is at its lowest resolution here, the bottleneck plays an important role in bridging the encoder and decoder paths.

3) Decoder: The decoder path mirrors the encoder, gradually reconstructing the spatial resolution of the segmentation map. This was achieved through transposed convolutional layers, which upsample the feature maps. At each level of upsampling, the decoder concatenated feature maps from the encoder through skip connections. These skip connections are vital to U-Net's success, as they restore spatial detail that would otherwise be lost during pooling operations in the encoder. By fusing contextual and fine-grained features, the decoder can delineate nuclear boundaries with higher accuracy.

4) Output Layer: Finally, the network concludes with a 1×1 convolutional layer with a sigmoid activation function. This produces a single-channel output in which each pixel value represents the probability of belonging to the nucleus class. A threshold of 0.5 was applied during evaluation to convert these probabilities into binary segmentation masks.

C. Training Process

1) Compilation: The model was compiled using the Adam optimizer and binary cross-entropy loss function. Since this was a binary segmentation task (nuclei vs. background), binary cross-entropy was suitable for training. The Dice coefficient was also used as an evaluation metric due to its relevance in segmentation tasks.

2) Callbacks: Early stopping was used to halt training if the validation loss did not improve for four consecutive epochs, thus preventing overfitting. Learning rate reduction was applied when the validation loss plateaued, reducing the learning rate by a factor of 0.5.

3) Training: The model was trained for 30 epochs, using the training dataset and evaluated on the validation dataset. The training process was optimized with the callbacks mentioned earlier to avoid overfitting and improve convergence.

D. Evaluation

Training the model required a loss function that could guide the network in distinguishing nuclei from background pixels, as well as evaluation metrics that could meaningfully capture segmentation quality. For this purpose, binary cross-entropy loss was chosen. Binary cross-entropy is widely used in binary segmentation tasks because it provides stable gradients and penalizes misclassifications at the pixel level. Its mathematical simplicity makes it efficient to optimize and well-suited as a baseline choice.

While binary cross-entropy is effective for optimization, it does not fully reflect segmentation quality in cases of class imbalance. In histopathology, nuclei typically occupy a small fraction of an image compared to the background, meaning that a model could achieve deceptively high accuracy by predicting mostly background. To address this, the Dice coefficient was used as the main evaluation metric. The Dice coefficient measures the overlap between predicted masks and ground truth annotations, providing a more reliable indicator of segmentation performance in imbalanced datasets.

In the current implementation, Dice was calculated after thresholding the predicted probabilities at 0.5. This allowed the metric to be reported in terms of binary masks. However, the use of "soft Dice," which computes the metric on continuous probability maps, could provide a smoother and more informative evaluation during training. In addition, future improvements could include using a combined loss function that incorporates both binary cross-entropy and Dice loss, thereby encouraging pixel-level accuracy while also optimizing region-level overlap. Alternative loss functions such as focal loss or Tversky loss could further help the model address difficult or small nuclei that are often underrepresented.

E. Optimization and Regularization

The model was optimized using the Adam optimizer with default learning rate parameters. Adam is a popular choice in biomedical segmentation tasks because it combines the benefits of adaptive learning rates and momentum, leading to faster convergence without requiring extensive manual tuning.

To prevent overfitting and improve convergence, several regularization strategies were employed through the use of callbacks. Early stopping was implemented by monitoring validation loss during training, which automatically halted training when the loss stopped improving for a set number of epochs. This ensured that the model did not continue training unnecessarily and restored the weights from the epoch with the best validation performance. In addition, a learning rate scheduler in the form of ReduceLROnPlateau was used. This callback decreased the learning rate when the validation loss plateaued, enabling the optimizer to make finer adjustments and potentially escape shallow minima.

Although these measures provided stability, additional regularization methods could be integrated in the future. Model checkpoints could be employed to automatically save the best-performing model based on validation Dice, ensuring reproducibility. Furthermore, setting random seeds across NumPy, TensorFlow, and Python environments would ensure consistent results across multiple runs. Techniques such as dropout within convolutional blocks or weight decay could also be explored to improve generalization.

F. Evaluation and Visualization

Evaluation of the model was conducted both quantitatively and qualitatively. From a quantitative perspective, the Dice coefficient served as the primary metric. A Dice score of 0.91 indicated that the model was able to achieve strong overlap between predicted masks and ground truth annotations. While this provided an encouraging benchmark, other metrics such as Intersection-over-Union (IoU), precision, and recall could have provided additional perspectives on performance. For instance, precision would highlight the rate of false positives, while recall would indicate the proportion of nuclei missed by the model.

Qualitative evaluation was equally important in this project. Predicted masks were visualized by overlaying them on the original histology images and comparing them to the ground truth annotations. In many cases, the U-Net successfully detected nuclei with accurate boundaries and consistent shapes. However, qualitative results also revealed failure cases. Overlapping nuclei were frequently merged into a single segmented region, reflecting the limitations of using binary masks rather than instance-level annotations. Additionally, faintly stained or very small nuclei were occasionally missed by the model.

To improve evaluation, post-processing methods such as connected component analysis could be introduced, allowing segmentation masks to be broken into individual objects for instance-level scoring. This would enable reporting of object-level F1 scores, which are more aligned with the biological applications of nuclei counting. Moreover, threshold sweeps could be performed to analyze how the Dice score changes with different binarization thresholds, providing further insight into model robustness.

IV. Results

The model was trained for 30 epochs, with performance monitored across multiple metrics: accuracy, Dice coefficient, and loss. These metrics were evaluated for both the training and validation datasets, providing insight into the model's segmentation capabilities.

The model showed substantial improvement in the Dice coefficient during the early epochs. Initially, the Dice coefficient increased from around 0.2 to 0.9 by Epoch 10, reflecting the model's effective learning. However, after Epoch 10, the improvements became minimal, suggesting that the model reached a plateau in its learning, as shown in Fig. 1.

Since Epoch 28 demonstrated the best performance (with 0.9130 Dice coefficient on the validation set and 0.0621 loss), the model's weights were restored from Epoch 28, ensuring the best performance was retained. This was done using the early stopping callback, which monitors validation loss and restores the model to the best epoch when validation performance starts to degrade.

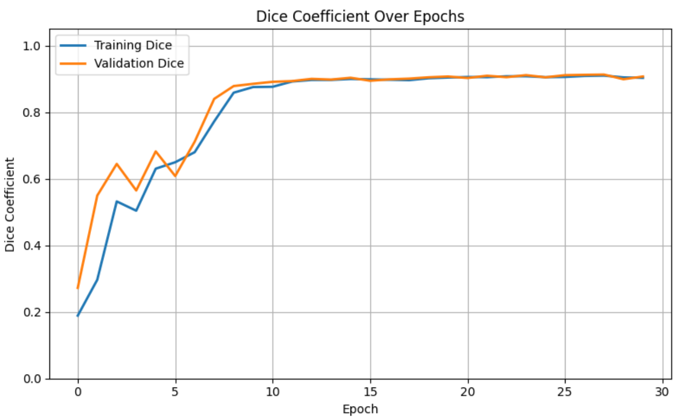

Fig. 1. Dice Coefficient over Epochs.

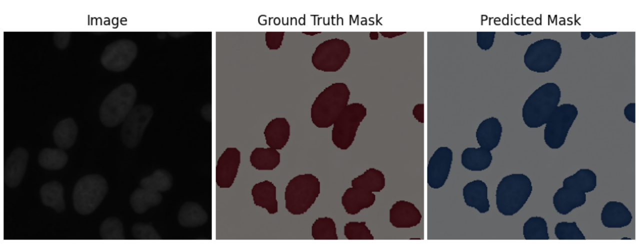

Fig. 2. Model Prediction of a WSI compared to the ground truth.

Visual inspection of segmentation outputs provided additional insights into the model's behavior. In many test cases, the U-Net successfully delineated nuclei boundaries and captured variations in nuclear shape and size. For example, elongated nuclei and irregularly stained nuclei were segmented with reasonable accuracy, demonstrating the network's capacity to learn morphological diversity. The segmentation masks aligned closely with the ground truth annotations, producing smooth and coherent nuclear boundaries.

However, the evaluation also revealed limitations. In images containing clusters of closely packed nuclei, the model frequently merged neighboring nuclei into a single region. This is an expected limitation of semantic segmentation, since the merging of instance masks into a binary mask during preprocessing removed information about individual nuclear boundaries. Another limitation observed was the tendency to miss faintly stained or very small nuclei, particularly when they blended into the background. These errors were consistent with known challenges in histopathology segmentation, where variability in staining and tissue preparation can obscure nuclear features.

Overall, the results confirmed that U-Net remains a robust baseline architecture for nuclei segmentation. The strong Dice performance suggests that the model effectively generalizes across diverse nuclear morphologies, but also highlights areas where architectural or methodological improvements are necessary for more precise results, especially in clustered or low-contrast regions.

V. Problems Faced

Although the project produced encouraging results, several significant challenges were encountered during development.

One of the most critical challenges was the impracticality of working with raw Whole Slide Images. WSIs are extremely large, often exceeding gigapixel resolutions, and cannot be fed directly into a deep learning model. A tiling strategy would have been required to split slides into smaller, manageable patches. Additionally, blank background regions of WSIs would need to be filtered out through tissue masking, and careful bookkeeping would be necessary to reconstruct predictions at the slide level. Beyond the technical considerations, the greatest challenge was annotation. Manually segmenting nuclei across thousands of tiles is prohibitively time-consuming, requiring expert knowledge and significant labor. This made it infeasible to build a high-quality WSI dataset within the timeframe of the project.

Another difficulty arose from the exploration of automated annotation tools. The Segment Anything Model (SAM), developed by Meta, was initially considered as a method for generating ground truth labels from WSIs. However, SAM proved unsuitable for this task. Due to the domain gap between natural images and histopathology images, SAM struggled to delineate small objects such as nuclei. Its predictions were often unstable, producing incomplete or noisy masks. Moreover, nuclei often occur in dense clusters, and SAM's instance segmentation approach frequently failed to separate them accurately. As a result, the output could not be used as reliable ground truth without extensive manual correction.

Another limitation of the current approach was the absence of stain normalization and data augmentation. Histopathology images are highly variable due to differences in staining protocols, scanners, and tissue preparation. Without stain normalization techniques such as Reinhard or Macenko normalization, the model is more vulnerable to overfitting to dataset-specific color distributions. Similarly, the lack of augmentations such as rotations, flips, and brightness adjustments reduced the diversity of training examples, limiting robustness to real-world variability.

Finally, the choice to merge all instance masks into a single semantic mask introduced structural limitations. While this simplification made training more tractable, it also meant that information about individual nuclei was lost. In practice, touching or overlapping nuclei were combined into single objects, which degraded the quality of segmentation in dense clusters. For downstream applications such as cell counting or nuclear morphology analysis, this merging poses a significant limitation.

VI. Future Work

There are several directions in which this project can be extended and improved. The most immediate need is the development of a full WSI tiling pipeline. Such a pipeline would involve splitting WSIs into overlapping tiles, applying tissue detection to exclude blank regions, and reassembling predictions into slide-level masks. Implementing multi-scale tiling strategies would further improve performance, as nuclei can vary significantly in size depending on magnification and tissue type. This would bring the approach closer to real-world pathology workflows.

Improving the quality of annotations is another critical area for future work. Instead of relying solely on public datasets, semi-supervised or weakly supervised approaches could be employed to reduce the need for exhaustive manual labeling. Active learning strategies could be particularly effective, where the model identifies uncertain regions and requests human annotation only where necessary, thus minimizing annotation costs. Another possibility is to fine-tune foundation models like SAM with domain-specific data, aligning them more closely with histopathology tasks.

From a modeling perspective, more advanced architectures could be investigated. Variants such as U-Net++ or Attention U-Net provide refined skip connections and attention mechanisms that may improve performance on challenging nuclear boundaries. Models such as HoVer-Net, which explicitly predict both nuclei masks and horizontal–vertical gradients for separating touching nuclei, offer promising avenues for overcoming the limitations of semantic segmentation. In addition, hybrid approaches that combine convolutional backbones with transformers may enhance the ability to capture both local and global dependencies.

The choice of loss function could also be improved. While binary cross-entropy worked as a baseline, a composite loss function combining cross-entropy and Dice loss would balance pixel-wise accuracy with region-level overlap. Focal loss or Tversky loss could further improve the detection of small or difficult nuclei, addressing the class imbalance problem inherent in this dataset.

Data preprocessing and augmentation remain areas of improvement as well. Implementing stain normalization techniques would standardize color distributions across slides, while stronger data augmentations could improve generalization across laboratories and staining protocols. Elastic deformations, in particular, would help the model adapt to morphological variability in tissue structure.

Finally, post-processing techniques should be incorporated to enhance instance-level performance. Watershed segmentation applied to probability maps could help separate overlapping nuclei. Connected component analysis could also be used to filter out small false-positive regions, thereby improving the quality of final segmentation masks.

Together, these improvements would transform the current prototype into a more robust and clinically applicable pipeline. By addressing dataset, architectural, and evaluation limitations, future work can build on the strong baseline established in this project to achieve state-of-the-art performance in nuclei segmentation for whole slide images.

VII. Conclusion

This project set out to address the challenge of cell nucleus segmentation in histopathology images, with a particular focus on whole slide image (WSI) analysis. Although the initial plan involved manually annotating raw WSIs, practical constraints related to image size, storage requirements, and annotation time made this approach infeasible within the project's scope. As a result, the study shifted to using publicly available, pre-segmented datasets, which enabled the implementation and evaluation of a reliable baseline model.

A U-Net architecture was developed and trained on these datasets, achieving a validation Dice coefficient of approximately 0.91. This result demonstrates the strength of U-Net as a baseline for nuclei segmentation, showing that the model can capture nuclear morphology with high accuracy under controlled conditions. The systematic evaluation revealed that the network is effective at detecting nuclei boundaries and segmenting diverse nuclear shapes, but also highlighted limitations in handling densely clustered or faintly stained nuclei.

Several practical challenges were encountered during the project, including the infeasibility of annotating WSIs, the unreliability of the Segment Anything Model for this domain. Addressing these challenges provided valuable lessons in data handling, model evaluation, and the importance of domain-specific adaptations in biomedical imaging.

Looking forward, the next steps toward building a clinically useful system include developing a complete WSI tiling pipeline, incorporating stain normalization and stronger augmentations, and adopting advanced architectures or loss functions to improve robustness. Instance-level post-processing methods such as watershed segmentation may also be required to accurately separate overlapping nuclei, which is essential for downstream applications such as cell counting and morphometric analysis.

In summary, this work demonstrates the feasibility of using U-Net as a strong baseline for nuclei segmentation and provides a foundation for future extensions aimed at WSI-scale deployment. By addressing current limitations, future iterations of this project could evolve into a robust and generalizable segmentation tool that contributes to automated pathology workflows and supports clinical decision-making.

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